



Figure 4: Twinkle Twinkle Little Star

which are really helpful to learn about new features and share your thoughts on music.

Getting started with MuseScore

In order to use MuseScore efficiently, you have to learn the notation system, which is a time-consuming process. However, let's just introduce ourselves to MuseScore by engraving the first two measures of 'Twinkle Twinkle Little Star'.

1. Start MuseScore Choose *File->New*. Give the title-like details, press the *Next* button, add *Piano* from the keyboards section and press the *Finish* button.
2. Now you have to enter the score shown in Figure 4. Press *N* on the keyboard to initiate note input. Choose the quarter note from the toolbar (or simply press *5* on your keyboard). Now you can enter the notes by clicking on the staff at the required positions (see the figure). For the last note of the second and fourth measures, you need to choose the half note from the toolbar (or press *6* on your keyboard). This is the same phase during which you use a physical MIDI keyboard for note entry rather than the mouse.
3. Press *N* again to switch off the note-entry mode. Now you can reposition (drag) or delete the notes to edit.
4. You can then press the *Play* button to play the score, or use *File->Save* to save or export it.

Tip: Save it as a MuseScore file for further editing. MIDI is best for a lightweight output, WAV for quality mixing/conversion and MP3/Ogg for distribution.

Troubleshooting

If MuseScore freezes when you hit the *Play* button, you can try changing the output settings from *Edit->Preferences*. The following configuration has been useful in Ubuntu.

Keep 'Use internal synthesiser' and PortAudio turned on (all others off). Under PortAudio, *Api=ALSA*. Change *Device* from the default setting to the actual hardware (an entry with codes like 'hw:0'). Press *OK* and restart MuseScore.

DAW applications

Digital Audio Workstation (DAW) is a facility that features the entire digital audio workflow — from recording to editing, mixing and mastering. It can be an electronic device or a computer. Needless to say, the second variant— software DAW — is the popular choice now. Let's have a quick look at some free DAW programs.

Rosegarden: Rosegarden is considered to be a leader in the GNU/Linux MIDI world. According to the Rosegarden website (rosegardenmusic.com), "Rosegarden is a music composition and editing environment based around a MIDI sequencer that features a rich understanding of music notations, and includes basic support for digital audio."

Rosegarden provides the classical score method, matrix editor and MIDI input for note entry (audio input is also available, but is not so popular). Recording the output of your MIDI device can easily be done using Rosegarden, and it seems to preserve the originality by considering even the velocity of a keypress.

Ardour: Ardour is a popular audio editor just like Audacity. It looks very professional, and it really is. While Audacity focuses on audio editing, Ardour aims to support the complete workflow from recording through mixing, editing and finally, mastering. Another key advantage of Ardour over Audacity is the enhanced support for MIDI. Audacity cannot even play MIDI while Ardour can treat it almost like an audio track.

LMMS: LMMS seems to be a 'new generation' music app that gives a lot of support to electronic music rather than to traditional instruments. There is a built-in wave synthesiser which can work out-of-the-box without any soundfont. However, it can work with soundfonts and sampled audio also. Samples can be loaded into a virtual keyboard and played as a given note with the help of automatic pitch shift.

LMMS means Linux MultiMedia Studio, though it is cross-platform now.

Other popular DAWs include Qtractor and Muse.

Audacity for further editing

Audacity could be the subject for a separate article. The free software project has become very powerful and has gained immense popularity.

The program can perform the A to Z of multi-track hard disk recording and features a large set of effects/plugin-ins. We will not go deep into Audacity, but merely have a 'get started' session.

Basic track management: You can import your audio clips using *File->Import->Audio* after starting a new project. Do not use *File->Import->MIDI* to import MIDI, but convert it first and import it as regular audio.

The track can be managed using the controls near each one. The selection tool from the toolbar can be used for cut-and-paste activities while the *Time Shift* tool helps us to shift the audio.

Recording: Anybody can record when there is a red button on the screen. However, make sure that you choose the right input device and the volume level is low (unamplified).

Amplification: Simply select the portion you need (or *Ctrl+A* to select all), and then choose *Effects->Amplify*. Sometimes you may want to turn on 'Allow clipping'. But keep in mind that it is not recommended.

Noise removal: Perhaps this is the most wanted feature. Thanks to Audacity, it is really simple.